SnapSplit

Intelligent Vision Extraction and Image Segmentation

Submitted By

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Introduction

SnapSplit is a web-based application designed to perform semantic segmentation on images, enabling users to identify and isolate objects within a picture. By generating pixel-level classification masks, it helps computers understand the structure and contents of an image in detail. This technology is valuable for a variety of applications, including autonomous driving, medical imaging, object detection, and AI-powered image analysis.

Problem Statement

The goal of this project is to build a machine learning model capable of automatically extracting the main subject from an image. For any given input picture, the model should output a new image in which only the subject is visible and everything else is rendered completely black. This subject isolation process can be used for automation in photography, digital art, augmented reality, virtual conferencing, and background replacement applications.

Use Cases: The project addresses one primary use case:

Automated Subject Isolation:

Description: Automatically detect and extract the main subject from any image. The output will be an image where only the subject is displayed as in the original photo, while the rest of the pixels are set to black. This replicates the "cutout" functionality required in many media editing pipelines.

Objectives

The main goal of the SnapSplit project is to provide an easy-to-use tool for object segmentation in images. By leveraging the DeepLabV3 deep learning model, the application aims to generate accurate segmentation masks that highlight objects of interest, making image analysis faster and more efficient. This project is designed for both researchers and enthusiasts working in computer vision and AI.

Key Objectives:

- Enable users to upload images (jpg, jpeg, png) for object extraction.
- Provide side-by-side visualization of the original and predicted segmentation.
- Allow users to download the segmented objects as PNG files.
- Make the interface interactive and user-friendly using Streamlit.

Overview

- SnapSplit is a deep learning-based image segmentation project that identifies and extracts objects from images using the DeepLabV3 model. The system handles data preparation, masking, augmentation, training, and inference, providing accurate pixel-level segmentation for objects of interest. Images are preprocessed, normalized, and augmented with techniques like flip, rotatio and stretching to improve model performance and generalization.
- The model is trained using PyTorch on a COCO dataset subset and fine-tuned. CrossEntropyLoss and Dice coefficient are used to optimize and evaluate the model. The workflow supports both CPU and GPU execution, with optional mixed-precision training for faster computation. Validation includes side-by-side comparisons of original images, ground truth masks, and predicted masks.
- The project is made user-friendly through a Streamlit interface, allowing users to upload images, view segmented results, and download extracted objects. It is extensible for future enhancements like multi-class segmentation, video processing, and integration into web or mobile applications, making it a practical tool for computer vision research, AI projects, and real-world object extraction tasks.

Dataset & Classes

- The project uses the COCO 2017 validation dataset (val2017) for training and evaluation. The full COCO dataset is very large, so val2017 provides a manageable subset of images while maintaining diversity in objects and scenes.

- Images are preprocessed by resizing to 255×255 or 256×256, normalizing pixel values, and generating binary masks for the target object class. Data augmentation is applied to improve model generalization.

- Currently, the model is trained for binary segmentation, classifying pixels as either background or object. For example, the “bear” category or “person” category can be extracted, depending on the task. This framework can be extended to multi-class segmentation in future enhancements.

Project Architecture

- The SnapSplit system follows a step-by-step workflow for object segmentation, starting from image input to output visualization. Users can upload images in jpg, jpeg, or png format via the Streamlit interface.

- Preprocessing & Augmentation: Uploaded images are resized, normalized, and optionally augmented with flips, rotations, brightness/contrast adjustments, and stretching to prepare them for the model. Masks are generated for the target object class.

- Segmentation using DeepLabV3: The preprocessed images are fed into the DeepLabV3 model with a ResNet-50 backbone, which predicts a segmentation mask classifying each pixel as object or background.

- Post-processing & Output: Predicted masks are applied to the original images to highlight objects of interest, and users can view side-by-side comparisons of original vs segmented images. Segmented objects can also be downloaded as PNG files for further use.

Key Features

- The SnapSplit application offers an interactive and user-friendly Streamlit interface, allowing users to easily upload images and view results.

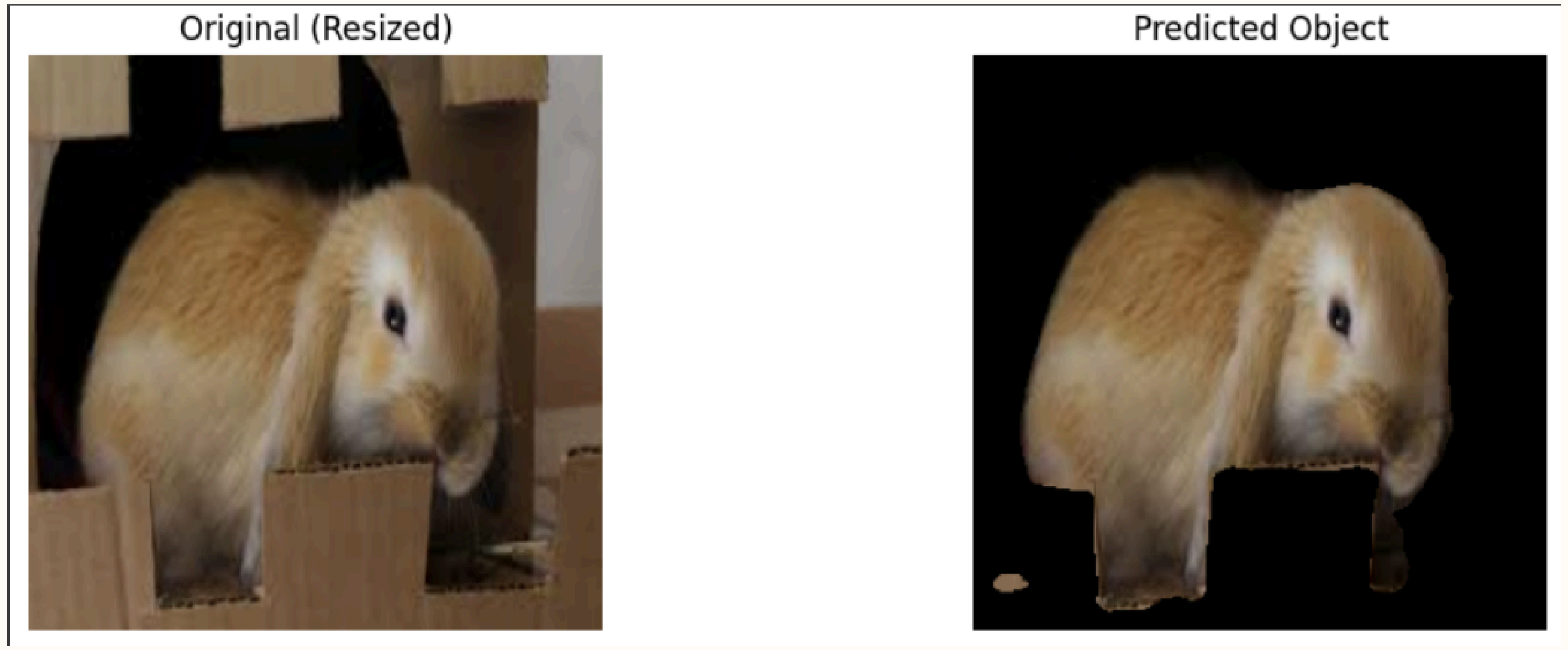
- Supports multiple image formats (jpg, jpeg, png) and provides side-by-side comparison of the original image and the predicted segmentation mask, making it easy to analyze results visually.

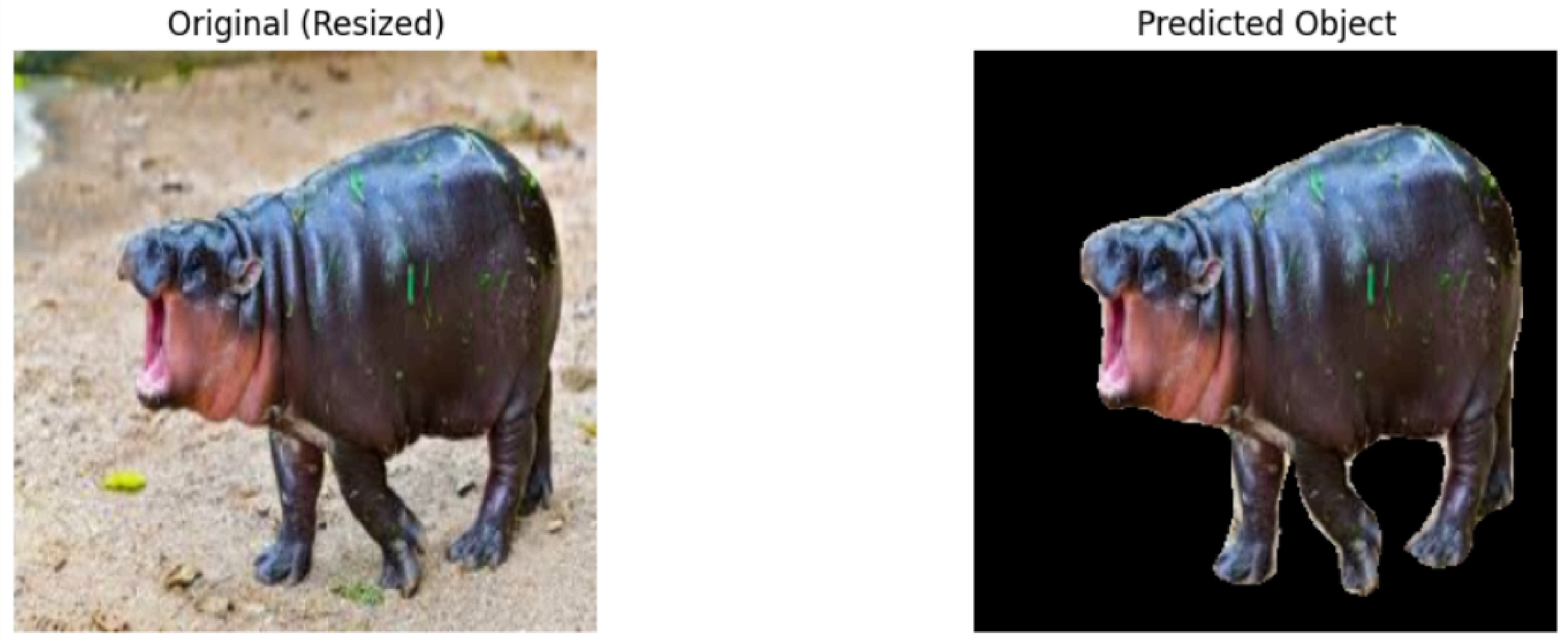
- Users can download the segmented objects as PNG files for further processing or use in other applications.

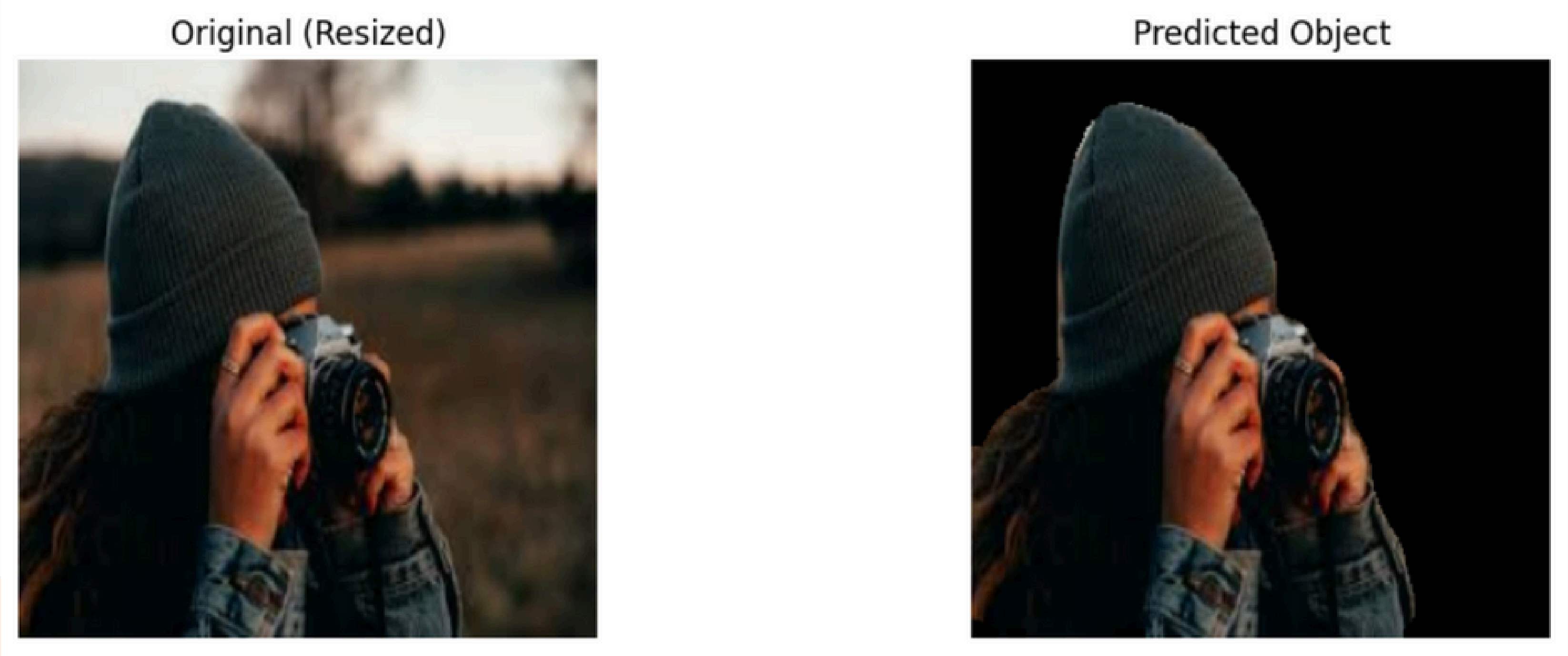
- The system is designed to run efficiently on both CPU and GPU, with support for optional mixed-precision training to speed up inference.

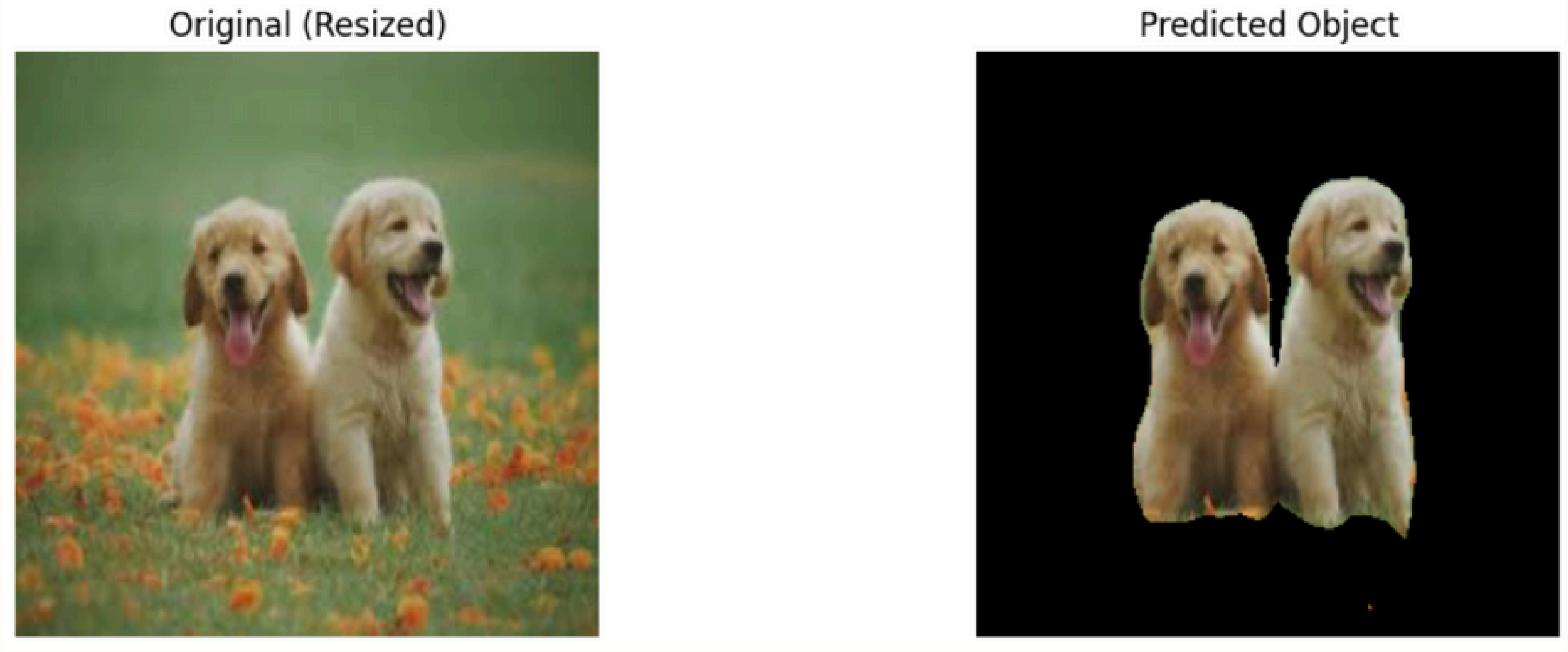
- Extensible for future enhancements, such as multi-class segmentation, video processing, and integration with web or mobile applications.

Sample Results

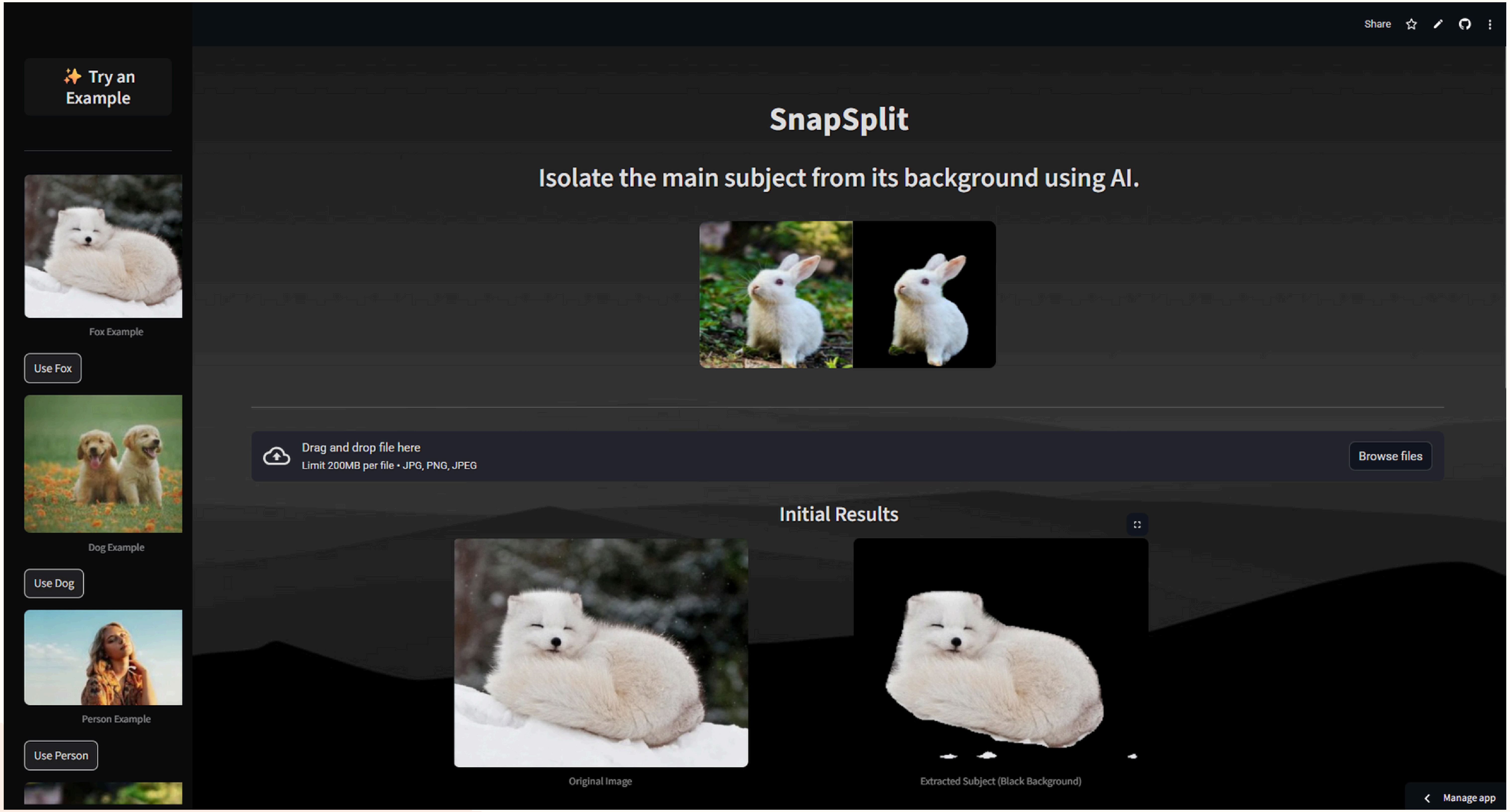








Sample Results



Advantages / Benefits

- Time-efficient and automated: SnapSplit eliminates the need for manual object masking, saving significant time while ensuring consistent results.

- High accuracy and reliability: Leveraging DeepLabV3, the system produces pixel-level segmentation masks with strong performance, validated through metrics like Dice coefficient and mIoU.

- User-friendly interface: The Streamlit app allows even non-technical users to upload images, view segmentation results, and download extracted objects effortlessly.

- Flexible and scalable: Works on CPU and GPU, supports multiple image formats, and can be extended for multi-class segmentation, video processing, or web/mobile deployment.

- Research and real-world applicability: Useful for computer vision projects, AI research, image analysis, and practical object extraction tasks.

Future Scope

- Real-time video segmentation: Adapt the system to process video streams, enabling object extraction in live or recorded videos for surveillance, robotics, or AR/VR applications.

- Web and mobile deployment: Integrate the model into web or mobile applications, allowing easy access and usage outside of development environments.

- Integration with AI pipelines: Combine segmentation with other AI tasks like object detection, tracking, or image enhancement for end-to-end computer vision solutions.
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- Handling large-scale datasets: Optimize for high-resolution images and larger datasets to improve model robustness and generalization in real-world scenarios.

Challenges & Solutions

- **Large dataset size:** The full COCO dataset is very large, making it difficult to handle locally.
- **Solution:** Used a manageable subset (val2017) for training and validation while maintaining diverse object samples.
- **Hardware limitations:** Personal laptop resources were insufficient for training deep learning models.
- **Solution:** Switched to Google Colab with GPU support, enabling faster training and efficient model experimentation.
- **Data variability and preprocessing:** Different image sizes, formats, and annotations posed challenges for consistent training.
- **Solution:** Applied resizing, normalization, and augmentation techniques to standardize inputs and improve model generalization.
- **Model optimization:** Training deep segmentation models can be slow and memory-intensive.
- **Solution:** Used batch training, mixed-precision computation, and efficient PyTorch DataLoaders to optimize performance.
- **Evaluation and visualization:** Comparing predicted masks to ground truth required careful processing.
- **Solution:** Implemented side-by-side visualization of original images, ground truth masks, and predicted outputs for clear evaluation.

Conclusion

The SnapSplit - Vision Extraction project successfully demonstrates accurate object segmentation in images using the DeepLabV3 model, providing precise pixel-level masks for objects of interest. The system is user-friendly, with an interactive Streamlit interface that allows easy image upload, visualization, and download of segmented objects. Through efficient preprocessing, augmentation, training, and inference, the project achieves reliable results even with hardware limitations by leveraging platforms like Google Colab. It is extensible for future enhancements, including multi-class segmentation, video processing, and web or mobile deployment, making it a practical tool for computer vision research, AI development, and real-world applications.

Thank you!